Paper Plain: Making Medical Research Papers Approachable to Healthcare Consumers with Natural Language Processing

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When seeking information not covered in patient-friendly documents, healthcare consumers may turn to the research literature. Reading medical papers, however, can be a challenging experience. To improve access to medical papers, we explore four features enabled by natural language processing: definitions of unfamiliar terms, in-situ plain language section summaries, a collection of key questions that guides readers to answering passages, and plain language summaries of those passages. We embody these features into a prototype system, Paper Plain. We evaluate Paper Plain, finding that participants who used the prototype system had an easier time reading research papers without a loss in paper comprehension compared to those who used a typical PDF reader. Altogether, the study results suggest that guiding readers to relevant passages and providing plain language summaries alongside the original paper content can make reading medical papers easier and give readers more confidence to approach these papers.

CCS Concepts: • **Human-centered computing → Interactive systems and tools**.

Additional Key Words and Phrases: augmented reading; plain language summaries; healthcare consumers

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1 INTRODUCTION

A robust public health system depends on the timely dissemination of medical findings to those who need them. Most often, people stay apprised of medical findings through conversation with their doctors, printed materials like pamphlets, and online resources like MedlinePlus or hospital websites [33, 57, 113]. However, these resources do not cover all medical conditions and treatments [12, 92], especially those which are the focus of emerging research [23, 88]. In many cases, the latest medical knowledge appears solely in the medical research literature [35, 39, 84, 103, 115]. For healthcare consumers such as patients, their families, and other caregivers, staying apprised of the latest research may mean becoming familiar with the literature. In the words of one patient [5]:

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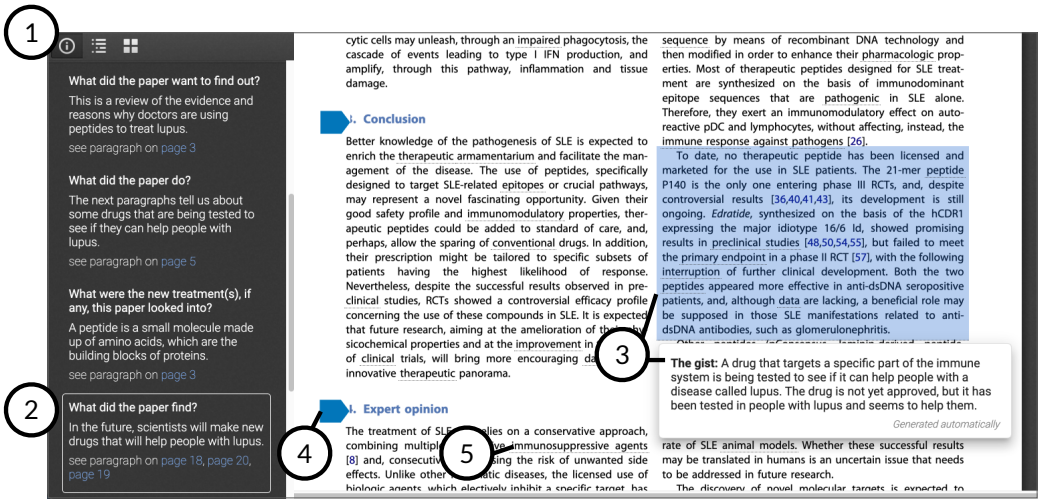


Fig. 1. Pictured is the Paper Plain user interface. When a reader opens a paper in Paper Plain, they see a side pane containing a reading guide (1), consisting of key questions the reader might ask of the paper, brief generated plain language answers, and pointers to passages in the paper where the reader can read more. When a reader clicks a question (2), the paper jumps to the passage that provides that answer and shows a paragraph-length plain language answer (3). Plain language summaries can be accessed for any section of the paper by clicking a label next to the section header (4). The reader can view definitions of medical terminology by clicking underlined terms (5).

> *I had been studying CLL [Chronic Lymphocytic Leukemia] through free access articles on PubMed and Google Scholar… Reading these NIH papers enabled me to have an intelligent dialogue with a CLL specialist, ultimately leading me to the selection of a clinical trial.*

For patients like the one quoted above, research articles offer an awareness of cutting edge medical findings and the nuance of underlying studies. Patients need not fully understand articles to derive some useful information. From these articles, patients may find and share information germane to treatment options with their healthcare providers [35, 84, 115].

However, a healthcare consumer's success in understanding the medical literature is by no means assured. Healthcare consumers report that, unsurprisingly, medical papers are difficult to read [35, 83]. This is in part due to being overwhelmed by the amount of unfamiliar terminology. It is also because healthcare consumers are unaccustomed to the norms of how research is conducted and how reports of it are structured [20, 35]. The result is that reading medical papers can be an experience that is challenging and at times demoralizing.

In this article, we ask how interactive information interfaces can make medical research articles approachable to non-expert healthcare consumers that need it, whom we refer to as "readers" in this paper. In particular, we study how articles can be imbued with new affordances to help readers navigate and evaluate their contents. The human-computer interaction literature demonstrates myriad ways that reading interfaces can assist readers, including by helping them understand unfamiliar terminology [8, 47], hiding sections that are predicted to be irrelevant [17], and answering user-written questions [114]. Drawing on this work as inspiration, we ask what combination of affordances would be necessary to help bridge the often enormous gap between a reader's current knowledge of biomedical research and a paper's contents. Consider, for instance, this sentence from a paper about systemic lupus erythematosus, linked from a patient-facing MedlinePlus page [101]: